

India's Agricultural Crop Production Analysis

TEAM ID	SWTID-2026-4021
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	1 MARK

5.4 – Team Management Tool for Agile Planning (Jira)

For the development of the Agricultural Crop Production Analysis System, an Agile project management tool like Jira is used to plan, track, and manage tasks efficiently. Jira helps in organizing work into sprints, assigning responsibilities, monitoring progress, and ensuring timely delivery of features such as crop analysis, yield prediction, and agricultural dashboards.

Why Jira for This Project?

- Supports Agile and Scrum methodology.
- Helps manage complex agricultural data system development.
- Enables sprint-wise planning and execution.
- Provides real-time project tracking and reporting.
- Improves collaboration among developers, analysts, and testers.
- Ensures transparency and accountability in the project.

Jira Project Structure

1. Epics

Large modules of the system:

- Crop Production Data Management
- Weather & Market Data Integration
- Yield Prediction System
- Dashboard & Visualization
- User Management System
- Reporting & Analytics

2. User Stories

Smaller functional requirements derived from epics:

- As a farmer, I want to view crop yield predictions.
- As an analyst, I want to analyze crop trends by region.
- As a policymaker, I want reports on agricultural productivity.
- As a user, I want weather-based farming suggestions.

3. Tasks & Sub-Tasks

Breakdown of work items:

- Develop data collection API
- Build machine learning model for yield prediction
- Design dashboard UI
- Integrate weather API
- Test analytics module

Agile Sprint Planning in Jira

Each sprint (2 weeks) is managed in Jira:

Sprint Workflow:

- To Do → Tasks planned for sprint
- In Progress → Tasks being worked on
- Code Review → Work under review
- Testing → QA validation phase
- Done → Completed tasks

Jira Features Used in the Project

1. Scrum Boards

- Visual representation of sprint tasks.

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- Easy tracking of progress.

2. Backlog Management

- All project requirements stored and prioritized.
- Tasks selected for each sprint.

3. Sprint Planning

- Assign tasks to team members.
- Estimate effort using story points.

4. Burndown Charts

- Track remaining work in each sprint.
- Helps ensure timely completion.

5. Reports & Dashboards

- Sprint reports
- Velocity charts
- Progress tracking dashboards

Team Roles in Jira

- Project Manager / Scrum Master
 - Sprint planning and monitoring
- Backend Developers
 - API and server-side development
- Frontend Developers
 - UI and dashboard development
- Data Analysts / ML Engineers
 - Crop prediction and analytics
- Testers (QA Team)
 - Testing and bug reporting
- DevOps Engineers
 - Deployment and cloud management

Benefits of Jira in This Project

- Improved coordination among teams.
- Clear task allocation and responsibility.
- Better visibility of project progress.
- Faster issue tracking and resolution.
- Efficient sprint-based delivery.
- Higher productivity and quality output.

Expected Outcome

Using Jira ensures structured Agile execution of the Agricultural Crop Production Analysis System. It helps in delivering modules like crop forecasting, data analytics, and dashboards in a timely and organized manner while maintaining high development quality and team collaboration.

Conclusion

Jira plays a crucial role in managing the Agile development of India's Agricultural Crop Production Analysis System. It provides a centralized platform for planning, tracking, and delivering project tasks efficiently, ensuring successful completion of the system with better coordination, transparency, and productivity.

BRAVO
— BUILDERS —
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